

SOC Attack Simulation & Detection Lab

Using Splunk Enterprise, Active Directory & Sysmon

Project Documentation

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Project Objective

This project demonstrates the implementation of a Security Operations Center (SOC) lab that simulates real-world cyber attacks within a Windows Active Directory environment. The lab integrates Active Directory, Sysmon, Splunk Universal Forwarder, and Splunk Enterprise to collect, monitor, analyze, and detect security events generated during attack simulations.

The project focuses on authentication monitoring, account management events, privilege escalation activities, and suspicious user behavior. Windows Security Event Logs and Sysmon logs are collected and forwarded to Splunk Enterprise, where they are analyzed using Search Processing Language (SPL). Interactive dashboards and real-time alerts enable continuous monitoring and rapid detection of suspicious activities, providing hands-on experience with enterprise SOC workflows.

Technologies Used

- Splunk Enterprise
- Splunk Universal Forwarder
- Microsoft Active Directory
- Sysmon
- Windows Server 2022
- Windows 10 Client

- Windows Security Event Logs
 - Search Processing Language (SPL)
 - VirtualBox
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Skills Demonstrated

- SIEM Implementation
 - Active Directory Administration
 - Windows Event Log Analysis
 - Sysmon Log Monitoring
 - SPL Query Development
 - Dashboard Creation
 - Alert Configuration
 - Threat Detection
 - Security Monitoring
 - Incident Investigation
 - MITRE ATT&CK Mapping
 - SOC Analyst Workflow
-

Repository Structure

SOC-Attack-Simulation-Detection-Lab-Splunk-Active-Directory/

```
|  
├── Architecture/  
├── Documentation/  
├── Queries/  
├── Reports/  
├── Screenshots/  
└── README.md
```

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Note: This project is developed for educational and portfolio purposes to demonstrate practical Security Operations Center (SOC) monitoring, attack simulation, log analysis, and threat detection using enterprise security technologies.

Page 2 – Acknowledgement

Acknowledgement

This project, **SOC Attack Simulation & Detection Lab using Splunk Enterprise, Active Directory & Sysmon**, was developed to gain practical experience in Security Operations Center (SOC) operations, Security Information and Event Management (SIEM), Windows security monitoring, and threat detection.

The primary objective of this project was to understand how cyber attacks can be simulated in a controlled Active Directory environment and how security events can be collected, analyzed, and investigated using enterprise security tools. Through this implementation, I gained hands-on experience with Windows Security Event Logs, Sysmon telemetry, Splunk Enterprise, Search Processing Language (SPL), dashboard creation, alert configuration, and incident investigation.

I would like to express my sincere gratitude to **Splunk Inc.** for providing Splunk Enterprise and Splunk Universal Forwarder, which made it possible to build this SIEM-based monitoring environment. I also acknowledge **Microsoft** for providing comprehensive documentation on Active Directory, Windows Security Event Logs, and Sysmon, which greatly supported the implementation of this project.

I am also thankful to the cybersecurity community, online learning platforms, official documentation, and virtual lab resources that helped me strengthen my understanding of SOC operations, Windows security monitoring, and threat detection through practical experimentation.

This project represents an important milestone in my journey toward becoming a **Security Operations Center (SOC) Analyst**. It reflects my commitment to continuous learning, practical skill development, and applying cybersecurity concepts in real-world scenarios.

I hope this documentation serves as a valuable learning resource for students, cybersecurity enthusiasts, and aspiring SOC Analysts who wish to understand how SIEM technologies can be used to detect, investigate, and respond to security events within a Windows Active Directory environment.

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Chapter 1: Project Overview

1.1 Introduction

Organizations today face an increasing number of cyber threats targeting Windows-based environments, particularly those using Active Directory for identity and access management. Security Operations Centers (SOCs) rely on Security Information and Event Management (SIEM) platforms to continuously collect, analyze, and correlate security events in order to detect malicious activities before they impact business operations.

This project, **SOC Attack Simulation & Detection Lab using Splunk Enterprise, Active Directory & Sysmon**, demonstrates the implementation of a practical SOC environment capable of monitoring, detecting, and investigating security events generated within a Windows domain. The lab simulates common attack scenarios, including failed login attempts, brute-force attacks, user account creation, password resets, and privileged group modifications.

Windows Security Event Logs and Sysmon telemetry are collected using Splunk Universal Forwarder and forwarded to Splunk Enterprise for centralized analysis. Using Search Processing Language (SPL), dashboards and real-time alerts are created to help security analysts identify suspicious behavior and investigate potential security incidents.

Rather than simply collecting logs, this project follows a complete SOC monitoring workflow—from attack simulation and log collection to threat detection, visualization, alerting, and incident investigation. It provides practical experience with technologies widely used in enterprise Security Operations Centers.

1.2 Problem Statement

Windows environments generate thousands of security events every day. These events include user authentication, account management, privilege changes, and system activities. Manually reviewing these logs through Windows Event Viewer is time-consuming, inefficient, and often results in important security events being overlooked.

Without centralized monitoring, organizations face challenges such as:

- Delayed detection of brute-force attacks.
- Limited visibility into user account modifications.
- Difficulty tracking privileged group membership changes.
- Slow investigation of authentication-related incidents.
- Lack of automated alerting for suspicious activities.
- Inability to correlate security events efficiently.

To address these challenges, organizations implement SIEM solutions that centralize log collection, automate analysis, generate alerts, and provide dashboards for continuous security monitoring.

1.3 Project Objectives

The primary objectives of this project are:

- Design and implement a Security Operations Center (SOC) lab.
 - Configure Microsoft Active Directory for user and group management.
 - Deploy Sysmon to generate enhanced endpoint telemetry.
 - Collect Windows Security Event Logs and Sysmon logs using Splunk Universal Forwarder.
 - Centralize log collection within Splunk Enterprise.
 - Simulate common attack scenarios in a controlled environment.
 - Develop SPL queries to analyze authentication and account management events.
 - Build interactive dashboards for continuous security monitoring.
 - Configure real-time alerts for suspicious activities.
 - Investigate security events using SOC analysis techniques.
 - Map detected activities to the MITRE ATT&CK Framework.
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1.4 Scope of the Project

This project focuses on implementing a practical SOC monitoring environment for Windows Active Directory infrastructures. The implementation includes log collection, attack simulation, event analysis, dashboard creation, and alert generation using Splunk Enterprise.

The project covers:

- Active Directory configuration
- Sysmon deployment
- Windows Security Event Log collection
- Splunk Universal Forwarder configuration
- Splunk Enterprise implementation
- Authentication monitoring
- Account management monitoring
- Privileged group monitoring
- Attack simulation
- Dashboard development
- Alert configuration

- Incident investigation
- Basic threat detection using SPL

The project does not include enterprise-scale SIEM clustering, endpoint response automation, SOAR integration, or advanced threat hunting across multiple network segments.

1.5 Expected Outcomes

Upon successful completion of this project, the following outcomes are achieved:

- Successfully deploy a functional Security Operations Center (SOC) lab using Splunk Enterprise, Active Directory, and Sysmon.
- Collect and centralize Windows Security Event Logs and Sysmon logs in Splunk.
- Simulate common cyber attack scenarios within a controlled Active Directory environment.
- Detect authentication attacks, account management activities, and privilege-related events using SPL queries.
- Build interactive dashboards for real-time security monitoring.
- Configure automated alerts for critical security events.
- Perform basic incident investigation using collected log data.
- Gain practical experience with enterprise SIEM workflows and SOC operations.

The completed lab demonstrates how security analysts monitor, detect, and investigate suspicious activities using industry-standard tools and techniques.

1.6 Technologies Used

Technology	Purpose
Splunk Enterprise	Central SIEM platform for log collection, searching, dashboards, and alerts
Splunk Universal Forwarder	Collects Windows Security and Sysmon logs and forwards them to Splunk
Microsoft Active Directory	Provides centralized user, group, and authentication management
Sysmon	Generates detailed endpoint telemetry for security monitoring
Windows Server 2022	Hosts the Active Directory Domain Services environment

Windows 10 Client	Simulates user activity and attack scenarios
Windows Security Event Logs	Primary source of authentication and account management events
Search Processing Language (SPL)	Used to search, analyze, and visualize collected logs
VirtualBox	Creates an isolated virtual lab environment

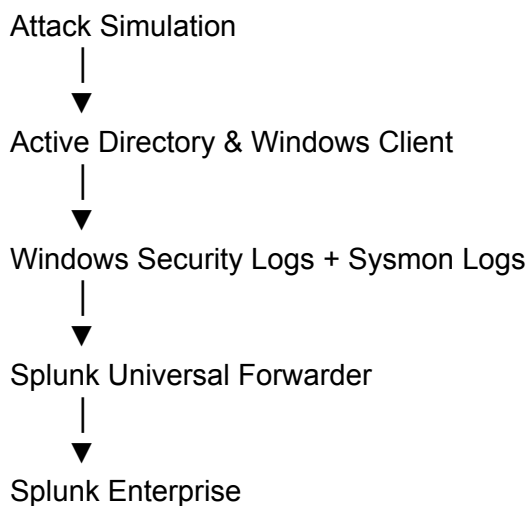
1.7 Project Workflow

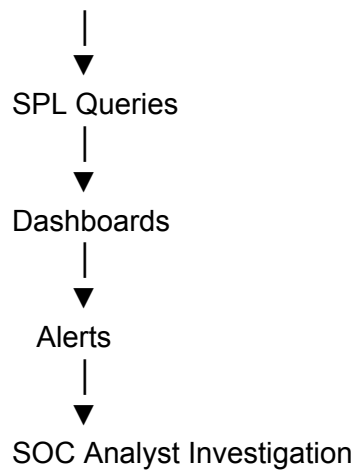
The project follows a structured SOC monitoring workflow from attack simulation to incident detection.

Workflow Steps

1. Configure Active Directory and Windows client machines.
 2. Install and configure Sysmon for enhanced event logging.
 3. Configure Splunk Universal Forwarder to collect Windows Security and Sysmon logs.
 4. Forward collected logs to Splunk Enterprise.
 5. Index security events for searching and analysis.
 6. Simulate security events such as failed logins, brute-force attempts, new user creation, password resets, and privileged group modifications.
 7. Analyze events using SPL queries.
 8. Build dashboards to visualize security activity.
 9. Configure alerts to detect suspicious behavior.
 10. Investigate alerts following SOC analyst workflows.
-

SOC Monitoring Workflow





Chapter Summary

This chapter introduced the objectives, scope, technologies, and workflow of the **SOC Attack Simulation & Detection Lab**. It explained the security challenges addressed by the project and outlined how Active Directory, Sysmon, Splunk Universal Forwarder, and Splunk Enterprise work together to provide centralized security monitoring, threat detection, and incident investigation.

The following chapter describes the complete laboratory architecture and explains the role of each component within the SOC environment.

Chapter 2: Lab Architecture

2.1 Introduction

A well-designed laboratory architecture is essential for understanding how security events are generated, collected, processed, and analyzed within a Security Operations Center (SOC). This project uses a virtual Windows-based environment to simulate real-world attack scenarios while enabling centralized monitoring through Splunk Enterprise.

The architecture integrates Microsoft Active Directory, Windows Security Event Logs, Sysmon, Splunk Universal Forwarder, and Splunk Enterprise to create a complete end-to-end detection workflow. Every security event generated during attack simulation is forwarded to the SIEM platform, where it is analyzed using SPL queries, visualized through dashboards, and monitored using automated alerts.

2.2 Lab Architecture Diagram

Figure 2.1 – SOC Attack Simulation & Detection Lab Architecture

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2.3 Architecture Components

The laboratory environment consists of the following major components:

1. Windows Client

The Windows client acts as the endpoint where normal user activities and simulated attacks are performed. Actions such as successful logins, failed login attempts, password resets, and user account modifications generate Windows Security Events and Sysmon telemetry.

2. Microsoft Active Directory

Active Directory provides centralized identity and access management. It authenticates users, manages groups, enforces security policies, and records account management activities such as user creation, account enablement, and privileged group modifications.

3. Sysmon

Sysmon extends native Windows logging by generating detailed endpoint telemetry, including process creation, network connections, and other security-relevant events. These logs provide additional visibility for threat detection and investigation.

4. Splunk Universal Forwarder

The Splunk Universal Forwarder continuously collects Windows Security Event Logs and Sysmon logs from the Windows system and securely forwards them to Splunk Enterprise for centralized analysis.

5. Splunk Enterprise

Splunk Enterprise serves as the central SIEM platform. It indexes incoming logs, enables powerful SPL searches, supports dashboard creation, generates alerts, and assists SOC analysts in investigating suspicious activities.

2.4 Data Flow

The following sequence describes how security events travel through the monitoring environment:

1. User activity or attack simulation generates Windows Security Events.
2. Sysmon records additional endpoint telemetry.
3. Splunk Universal Forwarder collects the generated logs.
4. Logs are securely forwarded to Splunk Enterprise.
5. Splunk indexes the received events.
6. SPL queries analyze authentication and security activities.
7. Dashboards visualize security events.
8. Alerts notify SOC analysts when suspicious behavior is detected.
9. SOC analysts investigate and validate the detected events.

2.5 Benefits of the Architecture

The architecture was designed to simulate a real-world Security Operations Center (SOC) environment while remaining simple, scalable, and easy to understand. By integrating Active

Directory, Sysmon, Splunk Universal Forwarder, and Splunk Enterprise, the lab provides end-to-end visibility into Windows security events.

Key benefits of this architecture include:

- Centralized collection of Windows Security Event Logs and Sysmon telemetry.
 - Real-time monitoring of authentication and account management activities.
 - Early detection of suspicious behavior through automated alerts.
 - Interactive dashboards for efficient security analysis.
 - Hands-on experience with enterprise SIEM workflows.
 - Safe environment for simulating cyber attacks without affecting production systems.
-

2.6 Design Considerations

Several design principles were followed while developing the SOC lab:

- **Modular Design:** Each component performs a specific function, making the environment easy to maintain and expand.
 - **Centralized Monitoring:** All security events are collected and analyzed from a single SIEM platform.
 - **Scalability:** Additional Windows endpoints can be integrated by deploying Splunk Universal Forwarder.
 - **Security Visibility:** Sysmon enhances Windows logging by providing detailed endpoint telemetry.
 - **Practical Learning:** The architecture closely resembles the workflow used in modern enterprise Security Operations Centers.
-

2.7 Architecture Summary

The architecture successfully demonstrates the complete lifecycle of security monitoring within a Windows Active Directory environment. Security events generated during attack simulations are collected through Splunk Universal Forwarder, enriched with Sysmon telemetry, indexed in Splunk Enterprise, analyzed using SPL queries, and presented through dashboards and alerts.

This implementation provides a practical understanding of how enterprise SOC teams collect, monitor, detect, and investigate security events using SIEM technology.

Chapter Summary

This chapter presented the complete laboratory architecture used throughout the project. It explained the purpose of each component, the end-to-end log flow, and the design considerations behind the SOC environment. Understanding this architecture provides the foundation for the implementation discussed in the following chapters.

The next chapter explains the laboratory environment, including the hardware, software, virtual machines, and network configuration used to build the SOC Attack Simulation & Detection Lab.

Chapter 3: Laboratory Setup

3.1 Introduction

Before implementing the SOC Attack Simulation & Detection Lab, a dedicated virtual environment was created to simulate a real-world Windows domain infrastructure. The objective was to safely generate security events, monitor system activities, and investigate simulated attacks without impacting a production environment.

The laboratory consists of Windows Server hosting Active Directory Domain Services (AD DS), a Windows client machine joined to the domain, Sysmon for enhanced endpoint monitoring, Splunk Universal Forwarder for log collection, and Splunk Enterprise as the centralized SIEM platform.

This isolated environment provides a practical platform for learning SOC operations, attack detection, log analysis, and incident investigation.

3.2 Lab Objectives

The laboratory environment was designed to achieve the following objectives:

- Create a secure environment for cyber attack simulation.
 - Configure Active Directory for centralized authentication.
 - Deploy Sysmon to generate enhanced security telemetry.
 - Collect Windows Security Event Logs and Sysmon logs.
 - Forward logs to Splunk Enterprise using Splunk Universal Forwarder.
 - Develop dashboards for security monitoring.
 - Configure alerts for suspicious activities.
 - Perform incident investigation using SIEM technologies.
-

3.3 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	16 GB Recommended
Storage	100 GB Free Disk Space

Operating System	Windows 10 / Windows 11 (64-bit)
Virtualization	Intel VT-x / AMD-V Enabled

These hardware specifications provide sufficient resources to run multiple virtual machines simultaneously while maintaining stable performance.

3.4 Software Requirements

Software	Purpose
Oracle VirtualBox	Virtualization Platform
Windows Server 2022	Active Directory Domain Controller
Windows 10 Client	Domain-Joined Workstation
Splunk Enterprise	SIEM Platform
Splunk Universal Forwarder	Log Collection Agent
Sysmon	Endpoint Monitoring
Windows Event Viewer	Security Event Verification
Web Browser	Access Splunk Web Interface

3.5 Virtual Laboratory Environment

The SOC Attack Simulation & Detection Lab was implemented within a virtualized environment to provide a safe and isolated platform for cybersecurity testing. Virtualization enables security events and attack scenarios to be simulated without affecting production systems.

The laboratory consists of a Windows Server configured as the Domain Controller, a Windows client joined to the Active Directory domain, and Splunk Enterprise used as the centralized SIEM platform. Sysmon was deployed on the monitored systems to generate enhanced endpoint telemetry, while Splunk Universal Forwarder continuously collected and forwarded security logs for analysis.

This virtual environment closely replicates the architecture used in enterprise Security Operations Centers (SOCs), providing practical experience with authentication monitoring, threat detection, and incident investigation.

3.6 Network Configuration

A virtual network was configured to allow secure communication between all systems within the laboratory. Proper network connectivity was verified before configuring log collection and monitoring services.

The communication workflow is summarized below:

1. Windows Client generates security events.
2. Active Directory authenticates users and records account management activities.
3. Sysmon captures additional endpoint telemetry.
4. Splunk Universal Forwarder collects Windows Security Event Logs and Sysmon logs.
5. Collected logs are forwarded to Splunk Enterprise.
6. Splunk indexes and analyzes the incoming data.
7. Dashboards and alerts provide visibility into security events.

This configuration ensures reliable log collection and centralized monitoring throughout the project.

3.7 Laboratory Workflow

The implementation followed a structured workflow to simulate real-world SOC operations.

1. Configure the Windows Server as the Active Directory Domain Controller.
 2. Join the Windows client to the Active Directory domain.
 3. Install and configure Sysmon.
 4. Deploy Splunk Universal Forwarder.
 5. Configure Splunk Enterprise to receive forwarded logs.
 6. Verify successful log collection and indexing.
 7. Simulate security events and attack scenarios.
 8. Analyze collected events using SPL queries.
 9. Create dashboards for continuous monitoring.
 10. Configure alerts to detect suspicious activities.
 11. Investigate generated alerts following SOC analyst procedures.
-

3.8 Benefits of the Laboratory Setup

The laboratory environment provides several advantages for cybersecurity learning and practical SOC training:

- Safe and isolated testing environment.
 - Hands-on experience with enterprise security technologies.
 - Practical understanding of Windows authentication monitoring.
 - Real-time log collection and analysis.
 - Experience with attack simulation and threat detection.
 - Improved incident investigation skills.
 - Reproducible environment for future testing and experimentation.
-

Chapter Summary

This chapter described the laboratory environment used to implement the SOC Attack Simulation & Detection Lab. It covered the hardware and software requirements, virtual environment, network configuration, and implementation workflow. The completed lab provides a realistic platform for simulating attacks, collecting security logs, and performing SOC monitoring using Splunk Enterprise, Active Directory, and Sysmon.

Chapter 4: Microsoft Active Directory

4.1 Introduction

Microsoft Active Directory (AD) is a centralized directory service used to manage users, computers, groups, and security policies within a Windows domain environment. It enables organizations to authenticate users, authorize access to network resources, and enforce security policies across multiple systems.

In modern enterprise environments, Active Directory plays a critical role in identity and access management (IAM). Since most authentication and account management activities occur within Active Directory, it is one of the primary targets for cyber attackers. As a result, monitoring Active Directory events is a fundamental responsibility of Security Operations Centers (SOCs).

In this project, Active Directory was configured to simulate real-world administrative and user activities. Security events generated by these activities were collected using Splunk Universal Forwarder, analyzed in Splunk Enterprise, and used to build dashboards and configure security alerts.

4.2 Role of Active Directory in This Project

Active Directory served as the central authentication and identity management system throughout the project. It was responsible for validating user logins, managing user accounts, controlling group memberships, and recording security-related events.

The following activities were performed within the Active Directory environment:

- User authentication.
- New user account creation.
- User account enablement.
- Password reset operations.
- Privileged group membership modifications.
- Security event generation for monitoring and analysis.

These activities generated Windows Security Event Logs, which were forwarded to Splunk Enterprise for centralized monitoring.

4.3 Active Directory Components Used

The project utilized the following Active Directory components:

Component	Purpose
Domain Controller	Authenticates users and manages Active Directory services.
Active Directory Users and Computers (ADUC)	Creates and manages users, groups, and organizational units.
Domain Users	Standard user accounts used for authentication testing.
Security Groups	Used to manage user permissions and simulate privilege changes.
Windows Security Logs	Record authentication and account management events.

4.4 Why Active Directory Monitoring is Important

Active Directory contains sensitive identity information and administrative privileges, making it a frequent target for cyber attacks. Continuous monitoring helps organizations quickly detect unauthorized activities and respond before attackers gain persistent access.

Monitoring Active Directory enables SOC analysts to:

- Detect unauthorized account creation.
- Monitor password reset activities.
- Identify privileged group modifications.
- Investigate authentication failures.
- Detect suspicious administrative actions.
- Improve visibility into identity-related security events.

4.5 Active Directory Events Monitored

To evaluate the effectiveness of the SOC lab, several Active Directory events were monitored throughout the project. These events represent common administrative activities and potential security risks frequently investigated by SOC analysts.

The following Windows Security Event IDs were monitored:

Event ID	Description	Purpose
4624	Successful Logon	Monitor successful user authentication
4625	Failed Logon	Detect failed login attempts and brute-force attacks
4720	User Account Created	Identify unauthorized account creation
4722	User Account Enabled	Monitor enabled user accounts
4724	Password Reset Attempt	Detect password reset activities
4728	Member Added to Security-Enabled Global Group	Monitor privileged group membership changes

These events were collected by Splunk Universal Forwarder and indexed into Splunk Enterprise for searching, dashboard visualization, and alert generation.

4.6 Active Directory Activities Performed

During the implementation of this project, several administrative tasks were performed within Active Directory to generate security events for monitoring and analysis.

The activities included:

- Creating new domain user accounts.
- Enabling disabled user accounts.
- Resetting user passwords.
- Adding users to privileged security groups.
- Performing successful and failed authentication attempts.
- Verifying that all generated events were successfully collected in Splunk Enterprise.

Each activity produced corresponding Windows Security Event Logs, allowing the detection rules and dashboards to be validated.

4.7 Security Monitoring of Active Directory

Active Directory security monitoring is essential because unauthorized changes to user accounts or privileged groups can lead to privilege escalation and unauthorized access.

Within this project, Splunk Enterprise was configured to continuously monitor Active Directory events and provide visibility into account management activities. Dashboards were created to display user account changes, while alerts were configured to notify the SOC analyst whenever critical events occurred.

This approach enables faster identification of suspicious administrative actions and supports timely incident response.

4.8 Benefits of Active Directory Monitoring

Monitoring Active Directory provides several security advantages:

- Improved visibility into user account activities.
- Early detection of unauthorized administrative actions.
- Monitoring of privileged group membership changes.
- Detection of suspicious authentication attempts.
- Support for incident investigation and forensic analysis.
- Centralized monitoring through Splunk Enterprise.

These capabilities strengthen the organization's security posture by enabling continuous monitoring of identity-related events.

Chapter Summary

This chapter explained the role of Microsoft Active Directory within the SOC Attack Simulation & Detection Lab. It described the key security events monitored during the project, the administrative activities performed to generate those events, and the importance of monitoring identity-related activities within a Security Operations Center.

The next chapter introduces **Sysmon**, explaining how it enhances Windows logging by providing detailed endpoint telemetry that complements Windows Security Event Logs.

Chapter 5: Sysmon

5.1 Introduction

Sysmon (System Monitor) is a Windows system service and device driver developed by Microsoft Sysinternals. It extends the default Windows logging capabilities by generating detailed endpoint telemetry that helps security analysts monitor system activities and detect malicious behavior.

Unlike standard Windows Security Event Logs, Sysmon provides deeper visibility into processes, network connections, file operations, and other endpoint activities. These events are valuable for threat detection, forensic investigations, and incident response.

In this project, Sysmon was deployed to enhance endpoint visibility and provide additional security telemetry alongside Windows Security Event Logs. The generated events were collected by Splunk Universal Forwarder and analyzed in Splunk Enterprise.

5.2 Role of Sysmon in This Project

Sysmon served as an endpoint monitoring tool that generated detailed security events during attack simulations. These logs complemented Windows Security Event Logs and provided additional context during security investigations.

Within this project, Sysmon was used to:

- Generate enhanced endpoint telemetry.
- Monitor system activities.
- Improve visibility into endpoint behavior.
- Support log analysis in Splunk Enterprise.
- Assist SOC analysts during incident investigation.

By combining Sysmon logs with Windows Security Event Logs, the project provided a more comprehensive view of system activities.

5.3 Why Sysmon is Important

Traditional Windows Security Logs focus primarily on authentication and account management events. While these logs are essential, they do not capture all endpoint activities.

Sysmon fills this gap by recording detailed information that helps analysts detect suspicious behavior that may not be visible through default Windows logging.

Using Sysmon provides several advantages:

- Enhanced endpoint visibility.
- Detailed process monitoring.
- Improved threat detection.
- Better forensic evidence.
- Support for incident investigations.
- Rich telemetry for SIEM analysis.

These capabilities make Sysmon one of the most widely used monitoring tools in enterprise SOC environments.

5.4 Sysmon Integration with Splunk

Within the laboratory environment, Sysmon events were continuously collected by Splunk Universal Forwarder and forwarded to Splunk Enterprise along with Windows Security Event Logs.

Once indexed, these events became searchable using SPL queries and could be correlated with authentication events generated by Active Directory.

This integration enabled centralized monitoring of both authentication activities and endpoint telemetry from a single SIEM platform.

5.5 Sysmon Events and Security Monitoring

During normal system operations, Sysmon continuously records endpoint activities and generates detailed event logs. These logs provide valuable information about processes, network activity, and system changes, enabling SOC analysts to identify suspicious behavior that may indicate malicious activity.

When combined with Windows Security Event Logs, Sysmon improves visibility into both authentication events and endpoint activities, making investigations more effective.

Within this project, Sysmon logs complemented authentication monitoring by providing additional context during attack simulations and security analysis.

5.6 Benefits of Using Sysmon

Deploying Sysmon within the SOC lab provides several operational and security advantages:

- Enhanced visibility into endpoint activities.
- Improved detection of suspicious system behavior.
- Detailed forensic evidence for incident investigations.
- Better correlation with Windows Security Event Logs.
- Continuous monitoring of endpoint events.
- Rich telemetry for SIEM-based threat detection.
- Faster investigation of suspicious activities.

These capabilities make Sysmon an essential tool for modern Security Operations Centers (SOCs).

5.7 Sysmon Workflow

The endpoint monitoring process implemented in this project follows the workflow below:

1. User activity or simulated attack occurs on the Windows system.
2. Sysmon records detailed endpoint events.
3. Splunk Universal Forwarder collects Sysmon logs.
4. Logs are forwarded to Splunk Enterprise.
5. Splunk indexes the received events.
6. SPL queries analyze the collected telemetry.
7. Dashboards and alerts provide visibility into suspicious activities.
8. SOC analysts investigate detected events using Splunk.

This workflow enables continuous endpoint monitoring and supports proactive threat detection.

5.8 Implementation Summary

Sysmon was successfully integrated into the SOC Attack Simulation & Detection Lab to enhance endpoint visibility and strengthen security monitoring. By generating detailed system telemetry and forwarding it to Splunk Enterprise, Sysmon complemented Windows Security Event Logs and improved the overall effectiveness of threat detection and incident investigation.

The combination of Sysmon, Active Directory, and Splunk Enterprise created a realistic SOC environment that demonstrates practical enterprise security monitoring.

Chapter Summary

This chapter explained the purpose and implementation of Sysmon within the SOC Attack Simulation & Detection Lab. It highlighted how Sysmon enhances native Windows logging by generating detailed endpoint telemetry and how these events are integrated with Splunk Enterprise to improve visibility, detection, and investigation of security incidents.

The next chapter introduces **Splunk Enterprise**, explaining its role as the central SIEM platform responsible for collecting, indexing, analyzing, and visualizing security events.

Chapter 6: Splunk Enterprise

6.1 Introduction

Splunk Enterprise is the central Security Information and Event Management (SIEM) platform used throughout this project. It collects, indexes, searches, and analyzes security logs generated by the Windows environment. By centralizing log management, Splunk enables SOC analysts to monitor authentication activities, investigate suspicious events, and respond to security incidents efficiently.

In this project, Splunk Enterprise receives Windows Security Event Logs and Sysmon logs from Splunk Universal Forwarder, where they are analyzed using Search Processing Language (SPL), visualized through dashboards, and monitored using automated alerts.

6.2 Role in This Project

Splunk Enterprise performs the following functions:

- Receives logs from Splunk Universal Forwarder.
 - Indexes Windows Security Event Logs and Sysmon events.
 - Executes SPL queries for log analysis.
 - Creates dashboards for real-time monitoring.
 - Generates alerts for suspicious activities.
 - Supports incident investigation and threat detection.
-

6.3 Data Processing Workflow

The monitoring workflow implemented in this project is:

1. Security events are generated on Windows systems.
 2. Splunk Universal Forwarder collects the logs.
 3. Logs are forwarded to Splunk Enterprise.
 4. Splunk indexes the events into the **main** index.
 5. SPL queries analyze authentication and account management events.
 6. Dashboards visualize security data.
 7. Alerts notify the SOC analyst of suspicious activities.
-

6.4 Key Features Used

Feature	Purpose
Data Indexing	Store Windows Security and Sysmon logs
Search & Reporting	Execute SPL queries
Dashboards	Visualize security events
Alerts	Detect suspicious activities
Search Processing Language (SPL)	Analyze collected logs

Implementation Summary

Splunk Enterprise successfully served as the central SIEM platform by collecting, indexing, analyzing, and visualizing security events generated during the attack simulations. Its dashboards and alerts enabled efficient monitoring and investigation of authentication and account management activities.

Chapter Summary

This chapter explained the role of Splunk Enterprise within the SOC Attack Simulation & Detection Lab. It demonstrated how centralized log collection, SPL analysis, dashboards, and automated alerts support continuous security monitoring and incident investigation.

The next chapter explains the deployment and configuration of **Splunk Universal Forwarder**, which is responsible for securely collecting and forwarding security logs to Splunk Enterprise.

Chapter 7: Splunk Universal Forwarder

7.1 Introduction

Splunk Universal Forwarder (UF) is a lightweight log collection agent that securely transfers data from source systems to Splunk Enterprise. It continuously monitors configured log sources and forwards new events in near real time while consuming minimal system resources.

In this project, Splunk Universal Forwarder was installed on the Windows Server and Windows client to collect Windows Security Event Logs and Sysmon logs and forward them to Splunk Enterprise for centralized monitoring.

7.2 Role in This Project

The Universal Forwarder acted as the communication bridge between the Windows environment and Splunk Enterprise.

Its responsibilities included:

- Collecting Windows Security Event Logs.
- Collecting Sysmon logs.
- Forwarding logs securely to Splunk Enterprise.
- Ensuring continuous log collection with minimal performance impact.
- Supporting real-time security monitoring.

Without the Universal Forwarder, security events would not be available for analysis within the SIEM platform.

7.3 Log Collection Workflow

The log forwarding process implemented in this project follows these steps:

1. Windows generates Security Event Logs and Sysmon events.
 2. Splunk Universal Forwarder continuously monitors the configured log sources.
 3. New events are securely forwarded to Splunk Enterprise.
 4. Splunk indexes the received logs.
 5. SOC analysts analyze the data using SPL queries, dashboards, and alerts.
-

7.4 Benefits of Universal Forwarder

The Universal Forwarder provides several operational benefits:

- Lightweight and efficient.
 - Continuous real-time log forwarding.
 - Secure communication with Splunk Enterprise.
 - Easy deployment and configuration.
 - Scalable for enterprise environments.
 - Reliable data collection with minimal resource usage.
-

Implementation Summary

Splunk Universal Forwarder was successfully configured to collect Windows Security Event Logs and Sysmon events from the monitored systems. The forwarded logs were indexed by Splunk Enterprise and made available for searching, dashboard visualization, and automated alert generation, enabling continuous security monitoring throughout the project.

Chapter Summary

This chapter explained the purpose and implementation of Splunk Universal Forwarder within the SOC Attack Simulation & Detection Lab. By continuously collecting and forwarding security logs, the Universal Forwarder enabled Splunk Enterprise to perform centralized log analysis, threat detection, and incident investigation.

The next chapter focuses on the **Windows Security Event IDs** monitored during the project and explains how each event contributes to threat detection and security monitoring.

Chapter 8: Windows Security Event IDs

8.1 Introduction

Windows Security Event IDs provide detailed information about authentication, account management, and administrative activities within a Windows environment. Monitoring these events enables SOC analysts to identify suspicious behavior, investigate security incidents, and detect potential attacks.

In this project, selected Event IDs were monitored using Splunk Enterprise to detect authentication failures, account modifications, password resets, and privileged group changes.

8.2 Event IDs Monitored

Event ID	Event Description	SOC Use Case
4624	Successful Logon	Monitor successful user authentication
4625	Failed Logon	Detect brute-force and failed login attempts
4720	User Account Created	Detect unauthorized account creation
4722	User Account Enabled	Monitor user account activation
4724	Password Reset Attempt	Detect password reset activities
4728	Member Added to Security-Enabled Global Group	Monitor privileged group membership changes

8.3 Detection Scenarios

The monitored Event IDs were used to detect the following security scenarios:

- Multiple failed login attempts indicating possible brute-force attacks.
- Successful user authentication after repeated failures.
- Creation of new user accounts.
- Enabling previously disabled user accounts.
- Password reset operations performed by administrators.
- Addition of users to privileged security groups.

These scenarios were visualized through dashboards and monitored using automated alerts.

8.4 Importance in SOC Operations

Monitoring Windows Security Event IDs provides valuable visibility into user authentication and administrative activities. These events help SOC analysts:

- Detect unauthorized access attempts.
 - Monitor identity-related changes.
 - Investigate suspicious account activity.
 - Identify privilege escalation attempts.
 - Support incident response and forensic investigations.
-

Implementation Summary

Windows Security Event IDs formed the foundation of the detection logic implemented in this project. By collecting and analyzing these events in Splunk Enterprise, the SOC lab successfully detected authentication failures, account management activities, and privileged group modifications through dashboards and real-time alerts.

Chapter Summary

This chapter explained the Windows Security Event IDs monitored throughout the project and their role in threat detection. These events served as the primary data source for dashboards, SPL queries, alerts, and incident investigations performed within the SOC Attack Simulation & Detection Lab.

The next chapter demonstrates how these events were generated through controlled **attack simulations** and how the resulting logs were analyzed using Splunk Enterprise.

Chapter 9: Attack Simulation

9.1 Introduction

To validate the effectiveness of the SOC monitoring environment, multiple attack scenarios were simulated within the Windows Active Directory domain. These activities generated real Windows Security Event Logs, which were collected by Splunk Universal Forwarder and analyzed in Splunk Enterprise.

The purpose of these simulations was to verify that dashboards, SPL queries, and alerts correctly detected and displayed security events in real time.

9.2 Attack Scenarios Performed

The following security scenarios were simulated during the project:

Attack Scenario	Event ID	Detection Purpose
Successful User Login	4624	Verify authentication monitoring
Failed Login Attempt	4625	Detect brute-force activity
New User Account Creation	4720	Detect unauthorized account creation
User Account Enabled	4722	Monitor account activation
Password Reset	4724	Detect password management activities
User Added to Privileged Group	4728	Monitor privilege escalation

9.3 Detection Process

Each simulated activity followed the same monitoring workflow:

1. An action was performed within the Active Directory environment.
2. Windows generated the corresponding Security Event.
3. Splunk Universal Forwarder collected the event.
4. The event was forwarded to Splunk Enterprise.
5. Splunk indexed the event into the **main** index.
6. SPL queries analyzed the event.
7. Dashboards displayed the activity.
8. Alerts were triggered for predefined detection rules.

This workflow confirmed that the monitoring environment was functioning correctly.

9.4 Project Outcome

All simulated attack scenarios were successfully detected by Splunk Enterprise. The generated events appeared in dashboards, matched the expected Event IDs, and triggered alerts where configured. This validated the effectiveness of the implemented SOC monitoring solution and demonstrated practical experience with attack simulation, log analysis, and security event detection.

Implementation Summary

The attack simulations successfully generated realistic Windows security events that were collected, analyzed, and visualized within Splunk Enterprise. The results confirmed that the configured dashboards and alerts accurately detected authentication events, account management activities, and privileged group modifications.

Chapter Summary

This chapter demonstrated how controlled attack simulations were used to validate the SOC monitoring environment. By generating real Windows Security Events and analyzing them in Splunk Enterprise, the project successfully showcased practical threat detection and security monitoring techniques.

The next chapter explains the **SPL queries** used to analyze these events and build the dashboards used throughout the project.

Chapter 10: SPL Queries

10.1 Introduction

Search Processing Language (SPL) is Splunk's query language used to search, filter, analyze, and visualize machine data. In this project, SPL queries were developed to detect authentication events, monitor account management activities, and support real-time security monitoring.

The queries transformed raw Windows Security Event Logs into meaningful insights that were used to build dashboards and configure alerts.

10.2 SPL Queries Implemented

The following SPL queries were developed during the project:

Query	Purpose
Successful Logins	Monitor user authentication (Event ID 4624)
Failed Logins	Detect failed authentication attempts (Event ID 4625)
Failed Login Trend	Visualize authentication failures over time
New User Created	Monitor account creation events (Event ID 4720)
User Enabled	Detect user account activation (Event ID 4722)
Password Reset	Monitor password reset activities (Event ID 4724)
Privileged Group Changes	Detect users added to privileged groups (Event ID 4728)

Latest Security Events Display the most recent security events

Top Event Codes Identify the most frequently occurring Event IDs

Note: The complete SPL queries are available in the project's [Queries/spl_queries.md](#) file.

10.3 Role of SPL Queries

The developed SPL queries enabled the SOC lab to:

- Search Windows Security Event Logs efficiently.
 - Detect suspicious authentication activities.
 - Monitor account management events.
 - Generate dashboard visualizations.
 - Trigger automated security alerts.
 - Support incident investigation.
-

Implementation Summary

The SPL queries successfully transformed Windows Security Event Logs into actionable security insights. They formed the foundation of the dashboards, alerts, and investigation workflows implemented within the SOC Attack Simulation & Detection Lab.

Chapter Summary

This chapter presented the SPL queries developed for the project and explained their role in monitoring authentication events and detecting suspicious activities. These queries enabled the creation of dashboards and alerting mechanisms that improved visibility into Windows security events.

The next chapter demonstrates how these queries were used to build interactive **Splunk dashboards** for real-time SOC monitoring.

Chapter 11: Dashboard Analysis

11.1 Introduction

Dashboards play a vital role in a Security Operations Center (SOC) by transforming raw security logs into meaningful visualizations. Instead of manually reviewing thousands of events, SOC analysts can quickly identify suspicious activities, authentication trends, and account management events through real-time dashboards.

In this project, multiple dashboards were created in Splunk Enterprise to monitor Windows Security Event Logs generated during the attack simulations.

11.2 Dashboards Created

The following dashboards were developed during the project:

Dashboard	Purpose
Successful Logins	Monitor successful user authentication
Failed Logins	Track failed authentication attempts
Failed Login Trend	Visualize failed logins over time
New User Created	Monitor newly created domain users
User Enabled	Track user account activation
Password Reset	Monitor password reset activities
Privileged Group Changes	Detect users added to privileged groups

Latest Security Events	Display recently generated security events
Top Event Codes	Analyze frequently occurring Event IDs

11.3 Dashboard Benefits

The dashboards provide several operational benefits:

- Centralized security monitoring.
 - Quick identification of suspicious activities.
 - Improved visibility into authentication events.
 - Faster incident investigation.
 - Continuous monitoring of Active Directory activities.
 - Simplified analysis of Windows Security Event Logs.
-

11.4 Dashboard Validation

Each dashboard was validated by generating the corresponding Windows Security Event. The dashboards updated successfully and accurately displayed the collected events, confirming that the monitoring environment was functioning as expected.

The dashboard visualizations demonstrated the successful integration of Active Directory, Sysmon, Splunk Universal Forwarder, and Splunk Enterprise.

Implementation Summary

The dashboards successfully transformed raw Windows Security Event Logs into meaningful visualizations, allowing security analysts to monitor authentication activities, account management events, and privilege-related changes in real time.

Chapter Summary

This chapter explained the dashboards developed within Splunk Enterprise and their role in security monitoring. By presenting authentication events and account management activities through interactive visualizations, the dashboards significantly improved visibility and supported faster threat detection and incident investigation.

The next chapter explains the **alert configuration and detection rules** implemented to automatically identify suspicious security events.

Chapter 12: Alert Configuration & Detection Rules

12.1 Introduction

While dashboards provide continuous visibility into security events, SOC analysts cannot monitor dashboards continuously. To improve detection capabilities, Splunk Enterprise was configured to generate automated alerts whenever predefined suspicious activities were detected.

The alerts implemented in this project help identify potential security threats in real time and support faster incident response.

12.2 Configured Alerts

The following alerts were implemented during the project:

Alert Name	Event ID	Detection Purpose
Brute Force Detection	4625	Detect excessive failed login attempts
New User Created	4720	Detect creation of new user accounts
Privileged Group Change	4728	Detect users added to privileged groups

Each alert was configured with appropriate search conditions and tested by generating the corresponding security events.

12.3 Detection Workflow

The alert generation process follows these steps:

1. A security event is generated on the Windows system.

2. Splunk Universal Forwarder forwards the event to Splunk Enterprise.
3. Splunk indexes and analyzes the event using SPL queries.
4. The configured detection rule evaluates the event.
5. If the rule conditions are met, an alert is triggered.
6. The SOC analyst reviews and investigates the alert.

This automated workflow reduces detection time and improves the efficiency of security monitoring.

12.4 Benefits of Automated Alerts

The implemented alerts provide several advantages:

- Early detection of suspicious activities.
 - Reduced manual monitoring effort.
 - Faster incident response.
 - Improved visibility into critical security events.
 - Support for proactive threat detection.
-

Implementation Summary

The configured alerts successfully detected brute-force login attempts, new user account creation, and privileged group modifications. The alerting mechanism demonstrated how Splunk Enterprise can automatically identify suspicious activities and assist SOC analysts in responding to potential security incidents.

Chapter Summary

This chapter explained the alerting mechanism implemented within the SOC Attack Simulation & Detection Lab. By automating the detection of critical Windows Security Events, the project demonstrated how SIEM platforms help SOC analysts identify threats quickly and initiate timely investigations.

The next chapter focuses on **Incident Investigation**, demonstrating how the generated alerts were analyzed and validated using Splunk Enterprise.

Chapter 13: Incident Investigation

13.1 Introduction

After an alert is generated, the next responsibility of a SOC analyst is to investigate the event and determine whether it represents legitimate activity or a potential security threat. During this project, each alert generated in Splunk Enterprise was reviewed by analyzing the corresponding Windows Security Event Logs and validating the associated user activity.

13.2 Investigation Process

The following investigation workflow was followed for each detected event:

- 1. Review the triggered alert in Splunk Enterprise.
- 2. Identify the associated Windows Event ID.
- 3. Verify the affected user account and host system.
- 4. Analyze related events using SPL queries.
- 5. Determine whether the activity is legitimate or suspicious.
- 6. Document the findings and recommended actions.

This structured approach ensures that security events are investigated consistently and efficiently.

13.3 Investigation Scenarios

Scenario	Investigation Performed
Failed Login Attempts	Reviewed failed authentication events and checked for repeated login failures.
New User Account Created	Verified whether the account creation was performed by an authorized administrator.
Password Reset	Confirmed the account involved and validated the password reset activity.

Privileged Group
Change

Verified the user added to the privileged group and confirmed whether the action was authorized.

Implementation Summary

The investigation process confirmed that the configured dashboards and alerts accurately reflected the simulated attack scenarios. By correlating Windows Security Event Logs with SPL search results, the project demonstrated a practical SOC investigation workflow for authentication and account management events.

Chapter Summary

This chapter demonstrated how security alerts generated by Splunk Enterprise were investigated using Windows Security Event Logs and SPL queries. The investigation process helped validate detection results and reflected the practical workflow followed by SOC analysts during security monitoring and incident response.

The next chapter maps the implemented detections to the **MITRE ATT&CK Framework**, providing industry-standard classification of the simulated attack techniques.

Chapter 14: MITRE ATT&CK Mapping

14.1 Introduction

The MITRE ATT&CK Framework is a globally recognized knowledge base that categorizes adversary tactics and techniques based on real-world cyber attacks. Security Operations Centers (SOCs) use this framework to understand attacker behavior, develop detection strategies, and improve incident response.

In this project, the simulated attack scenarios were mapped to relevant MITRE ATT&CK techniques to demonstrate how the implemented detections align with industry-standard threat detection practices.

14.2 Attack Mapping

Attack Scenario	Windows Event ID	MITRE ATT&CK Technique	Tactic
Failed Login Attempts (Brute Force)	4625	T1110 – Brute Force	Credential Access
Successful Login	4624	T1078 – Valid Accounts	Defense Evasion / Initial Access
New User Account Created	4720	T1136 – Create Account	Persistence
Password Reset	4724	T1098 – Account Manipulation	Persistence
User Added to Privileged Group	4728	T1098 – Account Manipulation	Privilege Escalation / Persistence

14.3 Benefits of MITRE ATT&CK Mapping

Mapping detections to the MITRE ATT&CK Framework provides several advantages:

- Standardizes detection rules using industry-recognized techniques.
 - Helps SOC analysts understand attacker behavior.
 - Improves threat hunting and incident response.
 - Enables better detection coverage analysis.
 - Aligns security monitoring with enterprise SOC practices.
-

Implementation Summary

The attack scenarios implemented in this project were successfully mapped to relevant MITRE ATT&CK techniques. This mapping demonstrates that the detection rules developed in Splunk Enterprise align with industry-standard threat detection methodologies and enhance the practical value of the SOC lab.

Chapter Summary

This chapter mapped the simulated attack scenarios to the MITRE ATT&CK Framework, demonstrating how authentication attacks, account creation, password resets, and privileged group modifications correspond to widely recognized adversary techniques. Incorporating MITRE ATT&CK enhances the project's relevance to modern SOC operations and enterprise threat detection.

The next chapter presents the **Testing & Validation** process used to verify that dashboards, alerts, and detection rules operated as expected.

Chapter 15: Testing & Validation

15.1 Introduction

After configuring the SOC Attack Simulation & Detection Lab, comprehensive testing was performed to verify that log collection, dashboards, SPL queries, and alerts were functioning correctly. Each simulated activity was validated to ensure that the corresponding Windows Security Event was successfully collected, indexed, and detected by Splunk Enterprise.

15.2 Test Scenarios

Test Scenario	Expected Result	Status
Successful Login (4624)	Event collected and displayed in dashboard	 Passed
Failed Login (4625)	Failed login detected and dashboard updated	 Passed
New User Created (4720)	User creation event detected	 Passed
User Enabled (4722)	Account enablement event detected	 Passed
Password Reset (4724)	Password reset activity recorded	 Passed
Privileged Group Change (4728)	Group modification detected	 Passed

Brute Force Alert

Alert triggered after threshold exceeded



Passed

15.3 Validation Results

The testing confirmed that:

- Windows Security Event Logs were successfully collected.
- Sysmon logs were forwarded correctly.
- Splunk Universal Forwarder operated without interruption.
- SPL queries returned accurate results.
- Dashboards displayed real-time security events.
- Alerts triggered as expected for configured detection rules.
- The complete monitoring workflow functioned successfully.

Implementation Summary

All components of the SOC lab were successfully validated through practical testing. The results confirmed that the implemented dashboards, SPL queries, and alerts accurately detected and visualized simulated security events, demonstrating the effectiveness of the monitoring solution.

Chapter Summary

This chapter presented the testing and validation process used to verify the SOC Attack Simulation & Detection Lab. Successful testing confirmed that log collection, event analysis, dashboards, and alerting mechanisms operated correctly, validating the overall implementation.

The next chapter discusses the challenges encountered during implementation and the troubleshooting steps taken to resolve them.

Chapter 16: Troubleshooting

16.1 Introduction

During the implementation of the SOC Attack Simulation & Detection Lab, several configuration and operational challenges were encountered. Resolving these issues provided valuable hands-on experience in SIEM deployment, Windows log collection, and Active Directory monitoring.

16.2 Issues Encountered

Issue	Cause	Resolution
Logs not appearing in Splunk	Incorrect Universal Forwarder configuration	Verified inputs, outputs, and restarted the forwarder service
No search results returned	Incorrect time range or missing Event ID	Adjusted the search time and generated new events
Dashboard panels not updating	No recent security events available	Generated fresh events and refreshed the dashboard
Alert not triggering	Detection threshold not reached	Simulated additional failed login attempts to trigger the alert
Missing Active Directory events	Activity not performed on the Domain Controller	Repeated the activity and verified event generation

16.3 Lessons Learned

The troubleshooting process reinforced several important practices:

- Verify log forwarding before creating dashboards.

- Confirm Event IDs are generated before testing SPL queries.
 - Always check the selected time range in Splunk searches.
 - Validate alert conditions using controlled attack simulations.
 - Test each component individually before integrating the complete solution.
-

Implementation Summary

Resolving these issues improved the stability and reliability of the SOC lab. The troubleshooting process strengthened practical skills in log analysis, SIEM configuration, and Windows security monitoring while ensuring accurate detection of simulated attack scenarios.

Chapter Summary

This chapter discussed the key challenges encountered during the implementation of the SOC Attack Simulation & Detection Lab and the steps taken to resolve them.

Troubleshooting these issues improved the overall reliability of the environment and provided practical experience with real-world SIEM administration and security monitoring.

The next chapter summarizes the technical and analytical skills gained through the successful completion of this project.

Chapter 17: Skills Learned

17.1 Introduction

The SOC Attack Simulation & Detection Lab provided practical experience with Security Information and Event Management (SIEM), Windows security monitoring, Active Directory administration, and threat detection. By implementing the complete monitoring workflow, this project strengthened both technical knowledge and analytical skills required for a Security Operations Center (SOC) Analyst.

17.2 Technical Skills Gained

During the implementation of this project, the following technical skills were developed:

- Splunk Enterprise Administration
 - Splunk Universal Forwarder Configuration
 - Active Directory User and Group Management
 - Windows Security Event Log Analysis
 - Sysmon Log Monitoring
 - Search Processing Language (SPL)
 - Dashboard Development
 - Alert Configuration
 - Security Event Investigation
 - SIEM Deployment and Monitoring
-

17.3 SOC Analyst Skills Developed

This project also strengthened practical SOC analyst capabilities, including:

- Authentication Monitoring
 - Log Analysis and Correlation
 - Threat Detection
 - Incident Investigation
 - Alert Validation
 - Security Event Monitoring
 - Account Activity Analysis
 - Privileged Access Monitoring
 - Basic Threat Hunting
 - Security Documentation
-

17.4 Project Achievements

The successful completion of this project demonstrates the ability to:

- Build a functional SOC monitoring lab.
 - Configure centralized log collection using Splunk.
 - Monitor Windows authentication and account management events.
 - Develop SPL queries for threat detection.
 - Create interactive security dashboards.
 - Configure automated detection alerts.
 - Investigate security events using SIEM tools.
 - Apply industry best practices for SOC monitoring.
-

Implementation Summary

This project provided hands-on experience with enterprise security tools and SOC workflows. It strengthened the practical skills required to collect, analyze, detect, and investigate Windows security events using Splunk Enterprise, Active Directory, and Sysmon.

Chapter Summary

This chapter highlighted the technical knowledge and practical SOC skills gained throughout the project. The implementation enhanced understanding of SIEM technologies, Windows security monitoring, threat detection, and incident investigation, providing a strong foundation for a career as a SOC Analyst.

The final chapter concludes the project by summarizing the implementation, outcomes, and key learning achievements.

Chapter 18: Conclusion

18.1 Project Summary

The **SOC Attack Simulation & Detection Lab using Splunk Enterprise, Active Directory, and Sysmon** successfully demonstrated the implementation of a practical Security Operations Center (SOC) monitoring environment. The project combined Windows Security Event Logs, Active Directory, Sysmon, Splunk Universal Forwarder, and Splunk Enterprise to simulate real-world security events and detect suspicious activities through dashboards and automated alerts.

Multiple attack scenarios, including failed login attempts, new user account creation, password resets, and privileged group modifications, were successfully simulated and monitored. The collected logs were analyzed using Search Processing Language (SPL), allowing security events to be visualized through dashboards and investigated using structured SOC workflows.

18.2 Project Outcomes

The project successfully achieved the following objectives:

- Implemented a functional SOC monitoring laboratory.
 - Configured centralized log collection using Splunk Enterprise.
 - Integrated Active Directory and Sysmon for enhanced visibility.
 - Monitored Windows authentication and account management events.
 - Developed SPL queries for security event analysis.
 - Built interactive dashboards for real-time monitoring.
 - Configured automated alerts for critical security events.
 - Performed incident investigation using collected security logs.
 - Applied MITRE ATT&CK mapping to simulated attack scenarios.
-

18.3 Key Learnings

This project provided valuable practical experience in:

- Security Information and Event Management (SIEM)
- Windows Security Event Log Analysis
- Active Directory Security Monitoring
- Sysmon Deployment and Monitoring
- Search Processing Language (SPL)
- Dashboard Creation
- Alert Configuration

- Incident Investigation
- Threat Detection
- SOC Operations

The hands-on implementation significantly improved the understanding of enterprise security monitoring and incident detection techniques.

18.4 Final Remarks

Building this SOC Attack Simulation & Detection Lab provided practical exposure to technologies and workflows commonly used in modern Security Operations Centers. The project demonstrates the ability to design a monitoring environment, collect and analyze security logs, detect suspicious activities, and investigate security events using enterprise SIEM tools.

The knowledge and experience gained through this implementation provide a strong foundation for future learning in threat hunting, incident response, malware analysis, and advanced security operations.

References

The following resources were used during the implementation of this project:

1. Splunk Enterprise Documentation
 2. Splunk Universal Forwarder Documentation
 3. Microsoft Learn – Active Directory Documentation
 4. Microsoft Learn – Windows Security Auditing
 5. Microsoft Sysinternals – Sysmon Documentation
 6. MITRE ATT&CK Framework
 7. Microsoft Windows Security Event ID Documentation
 8. Oracle VirtualBox Documentation
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Abbreviations

Abbreviation

Meaning

AD	Active Directory
AD DS	Active Directory Domain Services
SIEM	Security Information and Event Management
SOC	Security Operations Center
SPL	Search Processing Language
UF	Universal Forwarder
IAM	Identity and Access Management
ATT&CK	Adversarial Tactics, Techniques, and Common Knowledge